

FIXNET

API Design Specification

Detailed Endpoint Reference — v1

Document Version: 1.0

Status: Draft for Review

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Base Path: /v1

Related Document: FIXNET Authentication & Authorization Design (companion document)

1. Overview & Conventions

This document specifies all v1 REST API endpoints for the FIXNET platform, covering authentication, technician onboarding, messaging and quoting, payments and subscriptions, job lifecycle, and admin operations. Endpoints are grouped by domain, each with method, path, access control, request body, and success response examples.

Key conventions used throughout this specification:

- All monetary values are represented as integers in the smallest currency unit (RWF or cents) to avoid floating-point errors (REQ-PAY-002).
- All authenticated endpoints require a Bearer JWT in the Authorization header unless otherwise noted.
- Timestamps are ISO 8601 (UTC) unless otherwise specified.
- Access control roles are: customer, technician, support_agent, finance_admin, system_admin — see the companion Authentication & Authorization Design document for the full RBAC matrix.

2. Authentication & User Management — /v1/auth, /v1/users

1. Request SMS OTP

POST

/v1/auth/otp/request

Access: Public (rate-limited: 10 req/min per IP / phone number)

Request Body

```
{
  "phone_number": "+250788123456",
  "channel": "sms"
}
```

Success Response (200 OK)

```
{
  "status": "success",
  "message": "OTP sent successfully via SMS",
  "expires_in_seconds": 300
}
```

Error Response (429 Too Many Requests)

```
{
  "error_code": "RATE_LIMIT_EXCEEDED",
  "message": "Too many attempts. Locked out for 30 minutes."
}
```

2. Verify SMS OTP & Login**POST****/v1/auth/otp/verify***Access: Public***Request Body**

```
{
  "phone_number": "+250788123456",
  "code": "582910"
}
```

Success Response (200 OK)

```
{
  "access_token": "eyJhbGciOiJIUzI1Ni...",
  "refresh_token": "ref_a1b2c3d4e5f6...",
  "token_type": "Bearer",
  "expires_in": 900,
  "user": {
    "id": "a2c3b4d5-0000-1111-2222-333344445555",
    "phone_number": "+250788123456",
    "role": "customer",
    "preferred_language": "rw"
  }
}
```

3. Refresh Access Token**POST****/v1/auth/token/refresh***Access: Public***Request Body**

```
{
  "refresh_token": "ref_a1b2c3d4e5f6..."
}
```

Returns a new access_token and a rotated refresh_token; the previous refresh token is invalidated immediately.

4. Admin Portal Login & MFA Challenge

POST**/v1/auth/admin/login***Access: Public***Request Body**

```
{
  "email": "support@fixnet.rw",
  "password": "SecurePassword123!"
}
```

Success Response (202 Accepted — MFA Required)

```
{
  "mfa_required": true,
  "mfa_token": "mfa_temp_token_9988776655"
}
```

5. Admin TOTP Verification**POST****/v1/auth/admin/mfa-verify***Access: Public (requires valid mfa_token)***Request Body**

```
{
  "mfa_token": "mfa_temp_token_9988776655",
  "totp_code": "123456"
}
```

*Returns full access_token and refresh_token, identical in shape to the mobile OTP flow.***6. Get Current User Profile****GET****/v1/users/me***Access: Authenticated (all roles)***Success Response (200 OK)**

```
{
  "id": "a2c3b4d5-0000-1111-2222-333344445555",
  "phone_number": "+250788123456",
  "role": "technician",
  "preferred_language": "rw",
  "created_at": "2026-08-10T10:00:00Z"
}
```

7. Update Language Preference**PATCH****/v1/users/me/language***Access: Authenticated (all roles)***Request Body**

```
{
  "preferred_language": "en"
}
```

```
}

```

3. Technician Profiles & Onboarding — /v1/technicians

1. Register Technician Profile

POST /v1/technicians/register

Access: *Authenticated (technician role)*

Request Body

```
{
  "business_type": "individual",
  "service_categories": ["plumbing", "electrical"],
  "id_document_url": "s3://private-docs/id_12345.pdf"
}
```

Success Response (201 Created)

```
{
  "profile_id": "tech_profile_778899",
  "verification_status": "pending",
  "subscription_status": "lapsed",
  "message": "Registration submitted. Pending admin verification."
}
```

2. Search & Filter Technicians (Customer)

GET /v1/technicians

Access: *Authenticated (customer)*

Success Response (200 OK)

```
{
  "count": 12,
  "results": [
    {
      "id": "tech_profile_778899",
      "business_name": "Kigali Quick Fix",
      "service_categories": ["plumbing"],
      "rating": 4.8,
      "completed_jobs": 34,
      "is_topRated": true,
      "distance_km": 2.3
    }
  ]
}
```

Query params: *category=plumbing&lat=-1.9441&lng=30.0619&radius_km=10&page=1*

3. Fetch Single Technician Profile

GET /v1/technicians/{id}

Access: *Authenticated (customer)*

Returns profile details, uploaded portfolio content, star ratings, and review summaries.

4. Upload Work Portfolio or Spare Part Content

POST

/v1/technicians/me/content

Access: *Authenticated (technician)*

Request Body

```
{
  "type": "photo",
  "media_url": "https://storage.fixnet.rw/content/work_1.jpeg",
  "description": "Fixed leaking pipe under kitchen sink"
}
```

5. Fetch Technician Dashboard

GET

/v1/technicians/me/dashboard

Access: *Authenticated (technician)*

Success Response (200 OK)

```
{
  "trial_ends_at": "2026-09-10T00:00:00Z",
  "trial_days_remaining": 22,
  "leads_received_this_month": 18,
  "quotes_accepted": 12,
  "earnings_this_month_rwf": 150000
}
```

6. Complete Safety & Scam Training

POST

/v1/technicians/me/onboarding-complete

Access: *Authenticated (technician)*

Request Body

```
{ "training_completed": true }
```

4. Messaging, Quotes & AI Assistant — /v1/conversations, /v1/ai

1. Query Technician AI Assistant

POST

/v1/ai/chat

Access: *Authenticated (customer)*

Request Body

```
{
  "technician_id": "tech_profile_778899",
  "prompt": "Does this plumber handle copper pipes on weekends?"
}
```

```
}

```

Success Response (200 OK)

```
{
  "response": "Based on the technician's uploaded portfolio, they specify
    working on standard PVC and copper pipes. No explicit weekend hours
    are listed. Would you like to chat with them directly?",
  "confidence_score": 0.92,
  "can_answer": true
}
```

2. Initiate Chat Thread

POST
/v1/conversations

Access: *Authenticated (customer)*

Request Body

```
{
  "technician_id": "tech_profile_778899"
}
```

Success Response (201 Created)

```
{
  "conversation_id": "conv_11223344",
  "status": "chat_opened"
}
```

3. List User Conversations

GET
/v1/conversations

Access: *Authenticated (customer, technician)*

4. Get Conversation Messages

GET
/v1/conversations/{id}/messages

Access: *Participant / Support Agent*

Success Response (200 OK)

```
{
  "messages": [
    {
      "id": "msg_101",
      "sender_type": "customer",
      "content": "Hello, my water heater is leaking.",
      "is_masked": false,
      "created_at": "2026-08-23T10:15:00Z"
    },
    {
      "id": "msg_102",
      "sender_type": "technician",

```

```

    "content": "I can fix it today. My phone is [REDACTED]",
    "is_masked": true,
    "created_at": "2026-08-23T10:16:00Z"
  }
]
}

```

Query params: `limit=50&before_id=msg_99`

5. Send Structured Quote

POST `/v1/conversations/{id}/quotes`

Access: *Authenticated (technician)*

Request Body

```

{
  "total_price_rwf": 25000,
  "scope_notes": "Replacing heater valve and sealing joint"
}

```

Success Response (201 Created)

```

{
  "quote_id": "quote_556677",
  "status": "quote_sent"
}

```

6. Respond to Quote (Accept/Decline)

POST `/v1/quotes/{id}/respond`

Access: *Authenticated (customer)*

Request Body

```

{
  "action": "accept"
}

```

Success Response (200 OK)

```

{
  "status": "quote_accepted",
  "required_booking_deposit_rwf": 2500,
  "payment_next_step_url": "/v1/payments/booking-deposit"
}

```

5. Payments & Subscriptions — /v1/payments, /v1/subscriptions

1. Pay Booking Deposit

POST `/v1/payments/booking-deposit`

Access: *Authenticated (customer)*

Request Body

```
{
  "quote_id": "quote_556677",
  "payment_method": "momo",
  "phone_number": "+250788123456"
}
```

Success Response (202 Accepted)

```
{
  "transaction_id": "tx_momo_990011",
  "status": "pending",
  "user_instruction": "Approve the MoMo USSD prompt on your phone."
}
```

2. Payment Gateway Webhook**POST****/v1/payments/webhook***Access: Payment provider gateway (verified via signature header)***Request Body**

```
{
  "transaction_id": "tx_momo_990011",
  "quote_id": "quote_556677",
  "status": "successful",
  "amount_rwf": 2500
}
```

*Server action: atomically unlocks the technician's phone number in chat and changes job status to deposit_paid_unlocked.***3. Get Payment Receipt PDF****GET****/v1/payments/{id}/receipt***Access: Authenticated (payer customer)**Returns a PDF binary stream download.***4. Technician Subscription Payment****POST****/v1/subscriptions/subscribe***Access: Authenticated (technician)***Request Body**

```
{
  "plan_id": "plan_monthly_pro",
  "payment_method": "momo",
  "phone_number": "+250788123456"
}
```

6. Job Lifecycle & Service History — /v1/jobs

1. View Service History

GET

/v1/jobs/historyAccess: *Authenticated (customer)*

Success Response (200 OK)

```
{
  "jobs": [
    {
      "job_id": "job_001",
      "technician_name": "Jean-Pierre N.",
      "service_category": "plumbing",
      "date": "2026-08-15",
      "deposit_paid_rwf": 2500,
      "status": "completed"
    }
  ]
}
```

2. One-Tap Re-Book

POST

/v1/jobs/{id}/rebookAccess: *Authenticated (customer)**Creates or opens a direct chat with the former technician, bypassing search.*

3. Confirm Job Completion

POST

/v1/jobs/{id}/completeAccess: *Authenticated (customer)*

Request Body

```
{ "completed": true }
```

4. Submit Rating & Review

POST

/v1/jobs/{id}/reviewsAccess: *Authenticated (customer)*

Request Body

```
{
  "rating": 5,
  "comment": "Fast and clean plumbing work!",
  "photo_url": "https://storage.fixnet.rw/reviews/rev_1.jpeg"
}
```

7. Admin Portal Endpoints — /v1/admin

1. Verify or Reject Technician

PATCH `/v1/admin/technicians/{id}/verify`*Access: support_agent, system_admin***Request Body**

```
{
  "verification_status": "verified", // or "rejected"
  "rejection_reason": null
}
```

2. Moderate Uploaded Content**DELETE** `/v1/admin/content/{id}`*Access: support_agent, system_admin***Request Body**

```
{
  "reason": "Inappropriate content violation"
}
```

3. Resolve Dispute & Process Deposit Refund**POST** `/v1/admin/disputes/{id}/refund`*Access: support_agent, finance_admin, system_admin***Request Body**

```
{
  "refund_amount_rwf": 2500,
  "internal_notes": "Technician failed to show up for appointment."
}
```

4. Global Pricing Configuration**POST** `/v1/admin/config/pricing`*Access: system_admin***Request Body**

```
{
  "booking_deposit_type": "flat",
  "default_deposit_amount_rwf": 2000,
  "default_trial_days": 30
}
```

5. Generate Monthly Financial CSV Report**GET** `/v1/admin/reports/revenue`*Access: finance_admin, system_admin**Query params: month=2026-08&format=csv. Success response (200 OK) returns a CSV download (text/csv).***6. Account Suspension (With Audit Log)**

POST

`/v1/admin/users/{id}/suspend`

Access: *system_admin*

Request Body

```
{
  "reason": "Repeated off-platform payment scam reports"
}
```

Server action: locks the user profile and logs an entry into the immutable admin audit log (REQ-ADM-008).

8. Open Questions for Review

- Add/Remove Endpoints: are additional admin monitoring endpoints, SMS fallback hooks, or more granular technician categories needed?
- Modify Auth & Token Timers: should the 15-minute Access / 30-day Refresh token policies or rate limits be adjusted?
- Payload Structure Adjustments: are there custom fields needed in the JSON schemas for the mobile apps or the web portal?